



Review article

Nano-enabled improvements of growth and nutritional quality in food plants driven by rhizosphere processes



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ABSTRACT

With the rising global population growth and limitation of traditional agricultural technology, global crop production could not provide enough nutrients to assure adequate intake for all people. Nano-fertilizers and nano-pesticides have 20–30% higher efficacy than conventional products, which offer an effective solution to the above-mentioned problem. Rhizosphere is where plant roots, soil, and soil biota interact, and is the portal of nutrients transporting from soil into plants. The rhizosphere processes could modify the bioavailability of all nutrients and nanomaterials (NMs) before entering the food plants. However, to date, the overall rhizosphere processes regulating the behaviors and bioavailability of NMs to enhance the nutritional quality are still uncertain. In this review, a *meta-analysis* is conducted to quantitatively assess NMs-mediated changes in nutritional quality from food plants. Furthermore, the current knowledge and related mechanisms of the behavior and bioavailability of NMs driven by rhizosphere processes, e.g., root secretions, microbial and earthworm activities, are summarized. A series of rhizosphere processes can influence how NMs enter plants and change the biological responses, including signal transduction and nutrient absorption and transport. Moreover, future perspectives are presented to maximize the potentials of NMs applications for the enhancement of food crop production and global food security.

1. Introduction

Foods produced from the agricultural systems provide most of the nutrients that feed the world population, and the nutrient sufficiency is the basis of good health, productive lives, and longevity for everyone. Global population has reached 7.7 billion in 2020, continuously increasing to 9.7 billion in 2050 (United Nations, 2015). As a result, the total demand of agricultural production is expected to increase by up to 50%–80% (Lowry et al., 2019). However, the global cereal production is predicted to only reach up to 2.5 billion ton in 2020 and 3.1 billion ton in 2050 (Alexandratos and Bruinsma, 2012), which could not provide enough food to assure adequate nutrient intakes for all people (Ripple et al., 2019). Meanwhile, the efficiency of traditional agricultural technology has declined gradually since 2000, the flattening yield curves, decreasing crop resiliency, increasing environmental damage, and increasing global food demand have created an urgent need for new agricultural technologies (Lowry et al., 2019).

Due to the high ratio of surface-volume area, unique physico-chemical, mechanical and electronic properties in relation to their bulk counterparts (Naddeo et al., 2015), more and more engineered

nanomaterials (NMs), such as nano-fertilizer, nano-pesticides, and nano-sensors, have been successfully applied in agricultural production and management for the enhancement of soil nutrient bioavailability and food quality for the past 10 years (Kah et al., 2018; Scott et al., 2018). By 2020, the agricultural inputs using nanotechnology will reach \$75.8 billion, which will help promote global economic growth to \$3.4 trillion (Mishra et al., 2017). Kah et al. (2018) pointed out that the efficacy of nano-agrochemicals is up to 30% higher as compared to conventional products, nanotechnology could revolutionize the mode of agricultural production (Scott et al., 2018). However, despite the remarkable number of publications on the interaction of NMs with food plants, the knowledge regarding the modification of the nutritional value in edible food plant tissue by NMs is still limited.

Translocation of NMs into the edible food tissues and their possible impact on food quality and safety have been a major public concern (Zhao et al., 2014). Before entering the plants, NMs interact with soil components (e.g., minerals and organic matters), and thus their bioavailability could be modified by soil physico-chemical properties, particularly the pH and ionic strength of soil pore water (Tourinho et al., 2012). Also, large numbers of microbes (e.g., bacteria and fungi) and

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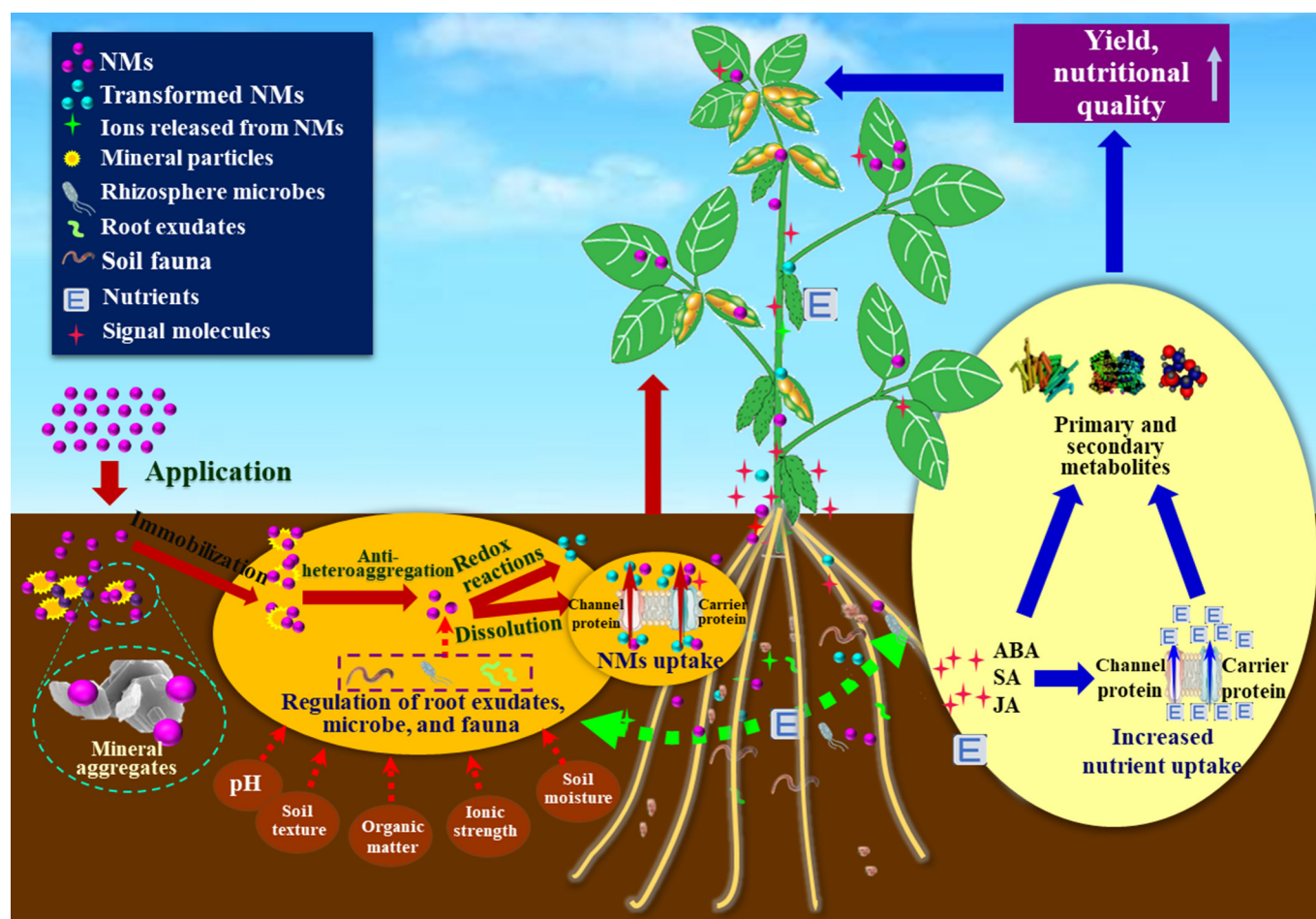


Fig. 1. Key environmental geochemical processes (left) and biological response mechanisms (right) of NMs in agroecological systems. Red solid arrow: key geochemical processes of NMs; Blue solid arrow: biological response of NMs; Green dashed arrow: the geochemical processes of NMs interactions with the biological response processes. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

soil fauna (e. g., earthworms) gather in the rhizosphere centering on plant roots, ultimately shaping a special biotic and abiotic environment through rhizosphere processes, e.g., root secretion, microbial growth, and earthworm activity in soils (Raaijmakers et al., 2009; Berendsen et al., 2012). These rhizosphere processes could alter the geochemical processes of NMs such as aggregation, anti-aggregation, redox reaction and transformation, and consequently, regulate the translocation of NMs into plants and the NMs-mediated physiological and biochemical response of edible crop tissue.

To date, however, the knowledge about the rhizosphere processes regulating geochemical behavior and bioavailability of NMs, and ultimately enhancing food nutritional quality is lacking. Thereby, we aim to assess the promoting effects of NMs on nutritional quality in food plants by *meta*-analysis, and further evaluate the main geochemical behaviors driven by rhizosphere processes (Fig. 1). This review will help to develop safe and sustainable nanotechnology for use in agroecosystems.

1.1. Overview of NMs-mediated effects on food plants

Plant growth parameters, e.g., germination, biomass, length, and yield, in response to different concentrations of CeO_2 , Cu, CuO, Fe_3O_4 and ZnO NMs, exhibit a trend of “low-promoting and high-inhibiting” after collecting and analyzing the data from the literature (Supporting Information) on the interaction between higher plants and NMs (Fig. 2). These specific NMs were chosen because that they were most widely used in agriculture and there are sufficient literature data to do the

statistical analysis and to find meaningful trends. The trends are not exactly the same for different types of NMs. For example, all the concentrations of CeO_2 NMs exposure lower than 100 ppm (11 data points) show significant promotion in plant growth as compared to un-exposed control (Fig. 2A, 2B). As the concentration range increases to 100–200 ppm, the overall plant growth is still significantly promoted, even with 16% of the 25 data points showing inhibition. In the concentration range of 200–500 ppm, the overall effect has no significant difference from control, with 67% of the 22 data points showing inhibition. As the concentration range increases to 500–1000 ppm and ≥ 1000 ppm, the plant growth is significantly inhibited, with 76% and 86% of the 28 and 50 data points showing inhibition, respectively (Fig. 2B). The overall significant promotions are exhibited in all Fe_3O_4 NMs concentrations as compared to control (Fig. 2C, 2D). Yet, as the concentration range increases, the extent of promotion decreases. 95%, 87%, and 67% of the 39, 15, 63 data points show significant promotions in the concentration range of < 20 ppm, 20–50 ppm, and ≥ 50 ppm, respectively. For Cu NMs exposure (Fig. 2E, 2F), plant growth displays significant promotion in the concentration range of < 1 ppm (100% of the 18 data points showing promotion) and 1–50 ppm (81% of the 16 data points showing promotion), but significant inhibition when more than 50 ppm (10% of the 20 data points showing promotion). CuO NMs show more inhibition than Cu NMs (Fig. 2G, 2H), the plant growth with a concentration range of < 10 ppm and 10–50 ppm CuO NMs has no significant difference with control, while the concentration range of 50–200 ppm and ≥ 200 ppm exhibits significant inhibition. For ZnO NMs, although some data points show inhibition, the overall effects are

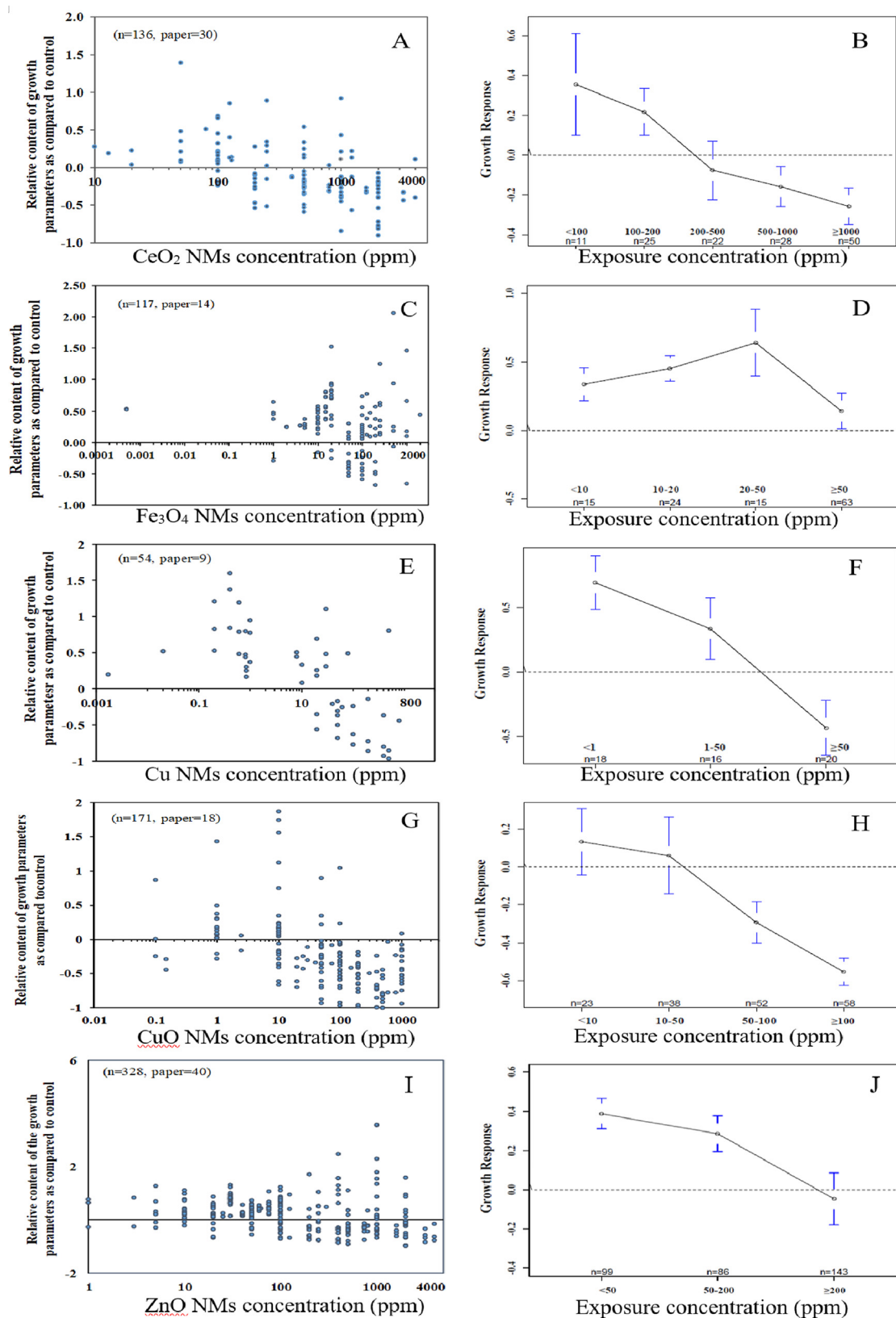


Fig. 2. The effect of CeO₂ NMs (A, B), Fe₃O₄ NMs (C, D), Cu NMs (E, F), CuO NMs (G, H), and ZnO NMs (I, J) on plant growth parameters (biomass, root/stem/shoot length, and yield). All the statistical data which have significant changes as compared to the control were collected from all NP-plant papers up to November 2019. Error bars in B, D, F, H, J denote 95% confidence intervals (CIs). n is sample size, paper is number of papers. The individual effect of NMs on plant growth response under each exposure concentration is significant if the 95% CI does not include zero.

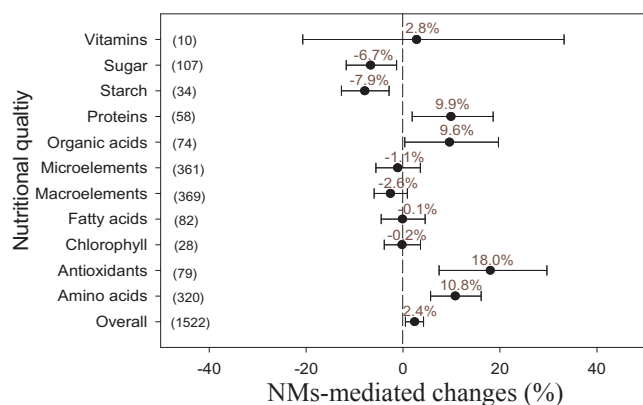


Fig. 3. The mean magnitude of NMs-mediated changes in on nutritional quality of food plants. Error bars denote 95% bias-corrected confidence intervals (CIs). Sample sizes are shown in brackets. The individual effect is significant if the 95% CI does not include zero.

significant promotion on plant growth in the concentration range of < 50 ppm and 50–200 ppm, but they have no significant difference when the concentration is over 200 ppm.

There are different methods of NMs applications in agricultural production. Our analysis shows that all the plant growth data upon foliar application of Fe_3O_4 NMs are significantly higher than control (Figure S1A, S1B). Meanwhile, inhibition appears at 20, 50, 100 ppm when using root exposure of Fe_3O_4 NMs (Figure S1C, S1D). In terms of the benefit and economic cost, appropriate foliar spray of Fe_3O_4 NMs seems relatively effective in agricultural practice. Moreover, the biological responses of plant growth at different stages upon exposure to NMs are different. In our analysis (Figure S1E–S1H), plant growth at both vegetative and reproductive stages upon 0–50 ppm Fe_3O_4 NMs application is significantly elevated. After the exposure concentration increases above 50 ppm, the plant growth at vegetative stage is significantly inhibited (100% and 56% of the 8 and 18 data points showing inhibition in 50–100 ppm and ≥ 100 ppm, respectively), while the plant growth at reproductive stage is still significantly elevated (100% of the 58 data points showing promotion).

After NMs exposure, the final quality of the edible part in plants is of major concern. 400 mg L^{-1} CeO_2 NMs promoted the content of starch, globulins and non-reducing sugars in cucumber fruits (Zhao et al., 2014). 100 mg kg^{-1} equivalent TiO_2 NMs exposure from soil and leaf notably increased antioxidant compounds lycopene in tomato fruits (Raliya et al., 2015). 250 mg kg^{-1} CeO_2 NMs did not change the accumulation of B and Na, but increased the content of nutrients (K, Ca, Mg, Zn, Fe, Cu) and amino acids (arginine and threonine) in barley grains (Rico et al., 2015). Application of 0.15–0.4 mg kg^{-1} Se^0 NMs to the soil improved Se content in the peach fruit (Bi et al., 2010). The levels of amino acids, proteins, and carbohydrates are analyzed (Figure S2). Amino acid content is significantly increased during the all NMs concentration ranges of < 500 ppm, 500–1000 ppm, and ≥ 1000 ppm as compared to control. The protein content exposed to NMs (< 200 ppm) and the carbohydrate content exposed to NMs (< 100 ppm) are significantly increased. However, as NMs concentration increases, no significant difference remained between control and NMs exposed food plants. The regulation of phytohormones and gene expressions could be intermediate factors or initial responses of plants to the NMs exposure (Yue et al., 2017). Plants at different levels (physiological, molecular, and metabolic levels) in response to metal/metal oxide NMs also exhibit the trend of “low-promoting and high-inhibiting”. As shown in Figure S2G, S2H, the phytohormone level in plants upon NMs (< 10 ppm) is significantly increased than control (53% of the 30 data points showing promotion). When the concentration increases over 10 ppm, the phytohormone level has no significant difference with control (81% of the 99 data points showing inhibition).

Total gene expressions are significantly enhanced (Figure S2I, S2J). Specifically, the expressions of genes related to antioxidation are significantly increased upon exposure of NMs. Except that the expressions of gene related to nutrient absorption exposed to 1000 ppm NMs and the gene related to metabolisms exposed to 100–200 ppm NMs have no significant differences with control, gene expressions related to antioxidation and metabolisms are significantly promoted (Figure S2K–S2P).

Overall, the effects of NMs on the growth of food plants vary greatly with plant species, NMs type, NMs modification, and NMs dose along with other factors (Rizwan et al., 2017). Optimal plant growth system could be established to tap the growth promoting potential of NMs in order to increase the efficiency of nanotechnology application in safe and sustainable agricultural systems. However, it is quite difficult to address all these influencing factors in one manipulative experiment of NMs-food plant interaction. Employing quantitative and heterogeneity explicated approach to analyze the various effects of NMs on food plant growth is a new research trend. Thus, a meta-analysis is further performed to quantitatively address the effects of NMs on the nutritional quality in food plant next section.

1.2. NMs-mediated nutritional quality of food plants: A meta-analysis

Meta-analysis is a quantitative synthesis method of research and can identify sources of variation in theme-based experimental results (Gurevitch et al., 2018). A meta-analysis is performed to quantitatively address the effects of NMs on nutritional quality, and further explore the potential impact factors. The criteria of the selected studies and methods of the meta-analysis referred to previous studies (Xiao et al., 2018; Chen et al., 2019), and are described in detail in the Supplementary Information. Study list and dataset used for the meta-analysis are shown in Table S1.

As shown in Fig. 3, NMs exposure increases overall nutritional quality of the edible part of food plants by 2.4% ($n = 1522$, $P = 0.012$). The “trim and fill” method detected 343 missing studies to the right of the overall mean. The addition of 343 missing cases to the dataset produced a new grand mean effect size of 15.2% (95% CIs 12.9% to 17.5%, $n = 1865$), suggesting a robust overall positive effect of NMs on nutritional quality of edible plants. The Rosenberg’s fail-safe numbers (Nfs) for the overall database is 620523, which is 81 times higher than the threshold of 7620 ($5 \times 1522 + 10$), also indicating a robust mean effect size. Specifically, upon NMs exposure, organic acids are significantly increased by 9.6% ($P = 0.042$), protein by 9.9% ($P = 0.030$), antioxidants by 18.0% ($P < 0.001$) and amino acids by 10.8% ($P < 0.001$), but sugar content is decreased by 7.9% ($P = 0.045$) (Fig. 3). However, there is no significant difference on the other nutritional quality indicators (e.g., chlorophyll, fatty acids, micro-elements, macro-elements, starch, and vitamins) between the un-exposed and NMs-exposed food plants (Fig. 3).

More specifically, the promoting effects of NMs on amino acids are notably influenced by NMs type ($Q_B = 15.78$, $df = 3$, $P = 0.001$) and plant type ($Q_B = 24.32$, $df = 4$, $P < 0.001$). NMs exposure increases the amino acid contents in barley and rice by 24.3% and 37.2%, respectively. CeO_2 and TiO_2 NMs increase amino acids by 14.3% and 26.8%, respectively (Table S2). Further, influence on amino acids content in food plants is more prominent in plants grown in soil matrix (Table S2) as the amino acids content increases by 10.7%, but not in plants grown in sand matrix. Besides, NMs exposure increases protein content in cucumber by 32.9%, ZnO NMs significantly increase protein content by 14.1%, and NMs only at the exposure dose of 500–1000 ppm increase protein content by 23.5% (Table S2). NMs-mediated increase in antioxidants is also dependent on NMs type ($Q_B = 37.27$, $df = 5$, $p < 0.001$), NMs modification ($Q_B = 12.15$, $df = 1$, $p < 0.001$), NMs dose ($Q_B = 36.07$, $df = 5$, $p < 0.001$) and plant type ($Q_B = 33.16$, $df = 4$, $p < 0.001$) (Table S2). However, no significant influence of NMs is observed on the overall micro- and macro-elements in food

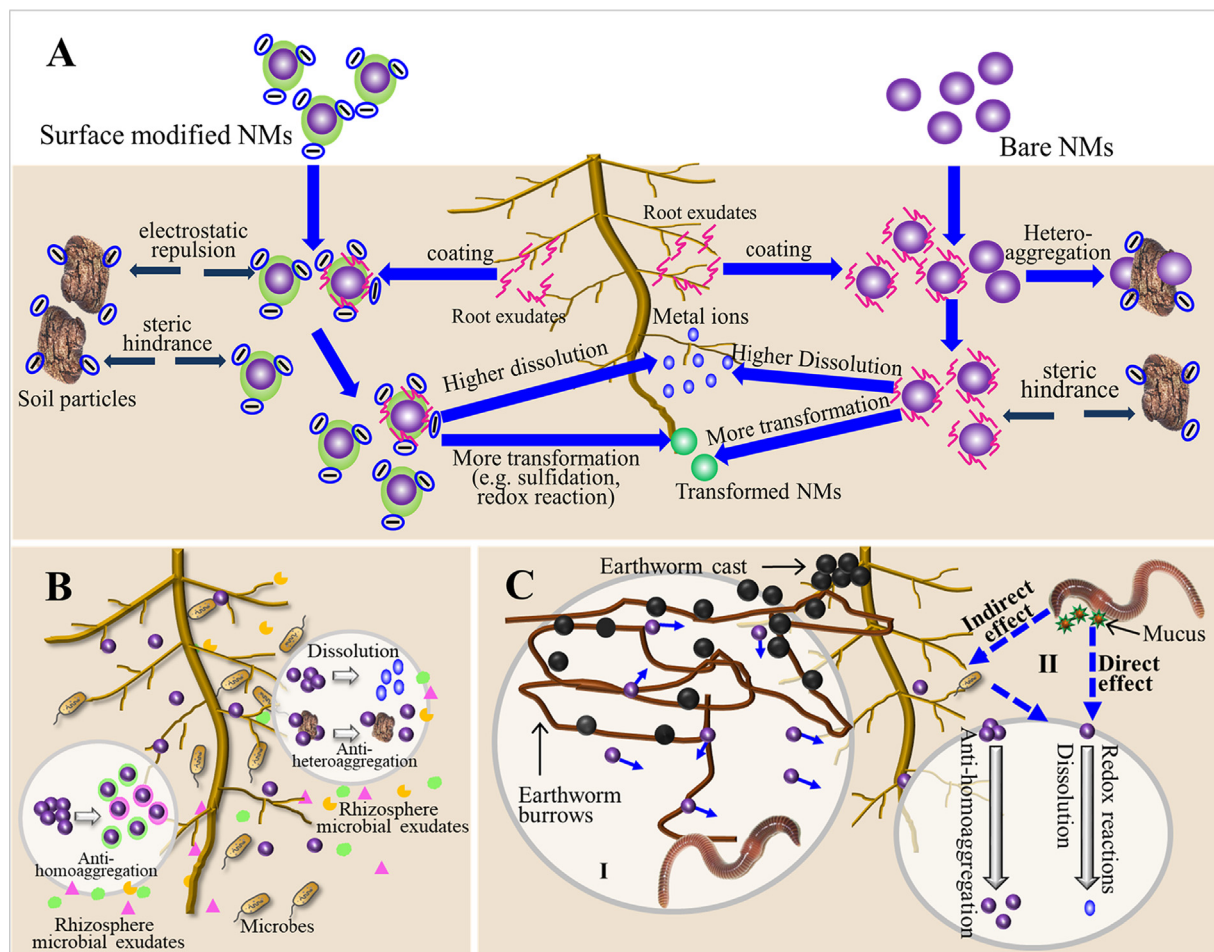


Fig. 4. Rhizosphere processes drive geochemical behavior and bioavailability of NMs. **A:** Effect of surface modification and root exudates on NMs bioavailability; **B:** Geochemical behavior and availability of NMs driven by rhizosphere microbes; **C:** The mobility and availability of NMs in rhizosphere as affected by earthworm bioturbation on the NMs mobility from bulk soils to rhizosphere, and direct and indirect impacts of earthworm mucus on NMs geochemical behaviors in rhizosphere, respectively.

plants. Even in some studies, for example, rice grains from CeO_2 NMs-treated plants had less Fe and S as compared to those from the untreated plants (Rico et al., 2013).

After NMs exposure, soil organic matter, and exudates from roots or rhizosphere microbes can adsorb on the surface of NMs and change the hydrophobicity, charge of NMs and/or transformation of NMs (Pettibone et al., 2008; Yan et al., 2008; Dimkpa et al., 2013a; Layet et al., 2017). Besides, the activities of soil fauna could also modify the physical and biochemical environment of rhizosphere soils (Bray et al., 2019). However, none of the studies used in our meta-analysis addressed the NMs-mediated differences in rhizosphere soil environment. To better use rhizosphere environment as a moderator, we further discuss how rhizosphere processes regulate the geochemical process and bioavailability of NMs in next section.

2. Geochemical behavior and bioavailability of NMs driven by rhizosphere processes

NMs could enhance the food plant yield and quality at relatively low concentrations. However, soil has large reactive surfaces where aggregations of NMs take place, which may decrease the mobility and bioavailability of NMs in soil (Batley et al., 2012), then inhibiting the increase of food plant yield and quality. NMs surface modification, rhizosphere fertilizer, and direct delivery to plant (e.g. inserting the roots in the emulsion bag with fertilizer and foliar sprays) are three main methods to increase NMs bioavailability against hetero-

aggregation. In this section, we focus on the surface modification and biological regulation of NMs by rhizosphere.

2.1. NMs surface modification

2.1.1. Decreasing hetero-aggregation between NMs with soil mineral particles

Hetero-aggregations of NMs arise from the capability of the two particles/surfaces to interact with each other, mainly via electrical force, van der Waals attraction force, hydrogen bonding, and/or chemical bonding (Wang et al., 2015). Therefore, the key against the hetero-aggregation of NMs with soil mineral particles is to control the particles surface interaction or to passivate the particle surface from being reacted “too quickly” by the stabilizer. Previous studies on stabilizer commonly focus on the electrostatic stabilizer (against the electrostatic attractive) and steric stabilizer (increase the steric hindrance) by the protocol and type of coating (Zhao et al., 2016a).

Negatively charged synthetic polymers, such as polyacrylic acid (PAA), carboxymethyl cellulose (CMC), and olefin maleic acid (OMA), could reduce the hetero-aggregation of NMs with negatively charged soil mineral particles by electrostatic repulsion (Schrick et al. 2004; He et al., 2007; Phenrat et al., 2010). Moreover, neutral molecules [e.g. polyacrylamide (PAM) and polyvinylpyrrolidone (PVP)] modified NMs, which could eliminate the electrostatic force and enhance the steric hindrance between NMs and mineral particles, have a good stability in soil (Basnet et al., 2013). PAA/Citrate-coating promoted the transport

and reduced the retention of silver NMs in soil, which is likely due to that PAA/citrate effectively blocking the available reactive sites on silver NMs for retention (He et al., 2019). In addition, some natural biopolymers (guar gum, xanthan gum, and starch) and organic matters (humic acid, citrate) have the same functions on NMs (Comba et al., 2011; Zhao et al., 2016a). However, there are still some constraints: 1) the concentration of stabilizer used for effective enhancement of NMs bioavailability is high; 2) the stabilizers might have negative effect on organisms. Therefore, there is still need for more effective and environment friendly stabilizers.

2.1.2. Increasing the interaction between NMs with plants

Surface coating can be used to increase the interaction of NMs with roots and promote the beneficial effects of NMs to plants. Avellan et al. (2019) demonstrated that PVP coating enhanced the uptake of Au NMs by wheat, due to enhanced hydrophobic interactions between the amphiphilic PVP coated Au NMs and plant surface. Citric acid-coated CeO₂ NMs increased the growth (indicated by shoot length) and pigment content of tomato (*Solanum lycopersicum* L.) at 210 days (Barrios et al., 2016). Well-dispersed carbon nanotubes (coated with -COOH and -PEG) increased the germination rate, biomass, water channel protein production in tomato, relative to poorly-dispersed bare carbon nanotubes (Villagarcia et al., 2012). Triethoxycaprylsilane modified ZnO NMs increased root and leaf length, and the nutriment elements (S, Mg) in kidney bean (*Phaseolus vulgaris*) when compared with bare ZnO NMs treatment (Medina-Velo et al., 2017). The direct delivery of NMs to plant is a high-efficiency pathway to enhance NMs bioavailability (Fig. 4A), surface-modified NMs could be used as rhizosphere fertilizer directly. However, the complex processes and direct exposure to highly concentrated NMs limit this type of application. Interestingly, several lines of evidence suggested the rhizosphere environment influenced by root exudates and soil organisms could alter bioavailability of NMs (Dimkpa et al., 2012).

2.2. Root exudates

Root exudate contains: low-molecular weight compounds including amino acids, organic acids, sugars, phenolics and secondary metabolites, and high-molecular weight compounds including mucilage and proteins (Bais et al., 2006). Root exudates could regulate the physico-chemical properties of rhizosphere solution (Layet et al., 2017; Zahra et al., 2017), further mediating homoaggregation and/or heteroaggregation of NMs in the rhizosphere (Chen, 2018; Shang et al., 2019). It was reported that the agglomeration of Ag NMs was reduced to minimal in root exudates (Dimkpa et al., 2013b). Heteroaggregation between NMs and soil particles was highly affected by root exudates (Ma et al., 2018). Wang et al. (2014) revealed that organic acids, one of the main root exudates, decreased the amount of micropores through dissolution of mineral particles and destruction of soil aggregates. Furthermore, the root exudates could affect the chemical processes (e.g. dissolution and transformation) of NMs in soil (Fig. 4A). The following four main points explain this scenario: 1) Root exudates could change the pH of rhizosphere, then affect the metal ions released from NMs. Zhao et al. (2016b) indicated that 10 mg L⁻¹ Cu NMs enhanced the root exudates concentration in hydroponic nutrient solution, and decreased the pH of solution. The increased proton concentration further enhanced the release of Cu ions. Zahra et al. (2015) also demonstrated that 50–200 mg kg⁻¹ TiO₂ and Fe₃O₄ NMs exposure decreased the pH of rhizosphere soil due to higher root exudates release. However, Gao et al. (2018) reported that CuO NMs (500 mg kg⁻¹) exposure increased the rhizosphere pH in weak acidic soil. This might be because that CuO NMs interacted with H⁺ from soil solution, releasing Cu²⁺ or Cu(OH)⁺ into the soil; the soil pH was gradually enhanced with consuming H⁺. The higher soil pH caused by CuO NMs could further decrease the dissolution rate and solubility of CuO NMs (Peng et al., 2017). 2) Root exudates could complex with NMs, and the root exudates-NMs

complexation could promote the release of Cu/Zn ions from CuO/ZnO NMs (Dimkpa et al., 2013a). However, the metal ions released from NMs are not directly related to the acidification of rhizosphere. 3) Root exudates could reduce Ce(IV) in CeO₂ NMs to Ce(III), and thus promote the uptake of Ce in plants (Zhang et al., 2019). IV) Root exudates could oxidize the release of ions from NMs into higher valence states. It was indicated that NMs would affect the formation of iron plaque after entering the soil. Organic acids from root exudates dissolved γ -Fe₂O₃ NMs and accelerated Fe ions release, and thus increasing the amorphous iron plaque formation (Pardo et al., 2016). However, the current studies on the interactions between NMs and root exudates were carried out under hydroponic or *in vitro* conditions, the electron donor or acceptor in root exudates are still unclear. The potential impact of soil environment, especially the organisms in rhizosphere soil, should be considered in future research.

2.3. Rhizosphere microbes

Rhizosphere soil contains millions of unique microbial species per gram soil (Curl and Truelove, 2012), owing to the ability to produce a large amount of bioactive secondary metabolites such as siderophores, lipopeptides and exopolysaccharides (Tyc et al., 2017). Rhizosphere microbes influence NMs bioavailability primarily through secreting such secondary compounds to mediate the dissolution and anti-aggregation of NMs, and thus could influence plant productivity (Fig. 4B) (Dimkpa et al., 2012).

Siderophores play a role in the chelation and dissolution of various elements and mineral phases (Chu et al., 2010; Hider and Kong, 2010). Previous studies found that the dissolution of hematite (Fe₂O₃) NMs was enhanced by elevated microbial siderophores production and the released Fe was further assimilated by the plant (Barton et al., 2012). Avellan et al. (2016) indicated that the metabolization of siderophore-mobilized Fe in Fe-doped high-aspect-ratio NMs (HARNs) had much less toxicity than that of HARNs. Siderophores can also chelate other metals such as Cu, Zn, Mn (Chen et al., 1994; Dimkpa et al., 2009; Brandel et al., 2012). Presumably, siderophores can potentially promote metal NMs dissolution and facilitate anti-heteroaggregation with soil minerals due to their high affinity for metal chelation (Neubauer et al., 2000).

The homo-aggregation of NMs is a major drawback in practical applications, which leads to the loss of active surface area and bioavailability reduction (Ocosoy et al., 2013). Lipopeptides as green bio-surfactants can be explored to enhance the stability of NMs for agricultural use (Sharma et al., 2009). Lipopeptides are amphiprotic compounds with surfactant activities produced by rhizosphere bacteria such as *Bacillus*, *Pseudomonas* and *Streptomyces* (Raaijmakers and Mazzola, 2012), which can disperse NMs and reduce the formation of aggregates by steric and depletion stabilization (Kiran et al., 2011; Rangarajan et al., 2018). *Bacillus* lipopeptides, as capping and stabilizing agent, resulted in more stable Ag NMs than naked Ag-NMs (Rangarajan et al., 2018). Similarly, the lipopeptides from *B. subtilis* composed of a mixture of the surfactin and fengycin, were also found to be an excellent dispersing agent for carbon NMs with high stability (Khan et al., 2011). Rhizosphere microbes also secrete variety of multi-functional exopolysaccharides (EPS), the adsorption of EPS on NMs could result in steric repulsion, which efficiently stabilized the NMs suspension (Khan et al., 2011; Lin et al., 2016). EPS derived from *Pseudomonas fluorescens*, a soil isolate from rhizosphere of periwinkle (*Catharanthus roseus*), could also produce stable Ag NMs that lasted for 4 months after which nano-silver started aggregating (Sirajunnisa and Surendhiran, 2014). Xiao et al. (2017) found that EPS of *Cordyceps sinensis* fungi prevented Se NMs from aggregating and improved their stability due to the strong interactions between Se NMs and the -OH groups of the EPS to form new C-O-Se bonds. In addition, the EPS-Se NMs exhibited enhanced antioxidant activities on superoxide anion radical (O₂^{•-}) and ABTS radical cation (ABTS^{•+}), which provided

potential application of EPS-Se NMs as nano-formulation for plant Se nutrition. Although, all these studies were performed in aqueous systems, the results highlighted the potential of EPS produced by rhizosphere bacteria to improve NMs stability through the surface coating, thus enhancing the bioavailability of NMs for plants. Other microbial processes, e.g., the excretion of organic acids, thiosulfate and cyanide, may also drive metal solubilization. Cyanide may play an important role in transforming the primary to secondary Au grains by providing mobile Au-complexes (Rea et al., 2016).

Overall, rhizosphere microbes have the potential to improve the biological activity of NMs mediated by dissolution, dispersion, and anti-aggregation (Fig. 4B). However, to what extent the association of NMs with rhizosphere microbial cells can enhance bioreactivity of NMs needs further confirmation, more *in planta* researches should be conducted. Given that solubilization and homo-aggregation of NMs are affected differently and simultaneously by various environmental factors such as pH, ionic strength, organic matter (Amde et al., 2017), it is difficult to predict the outcomes in the agricultural field. Nevertheless, knowledge of interactions between plant growth promoting rhizobacteria (PGPR) and NMs in the rhizosphere could provide promising opportunity to explore cost effective and environmentally friendly nano-formulations for agricultural applications. Recent studies showed that NMs have positive effects on PGPR by promoting the growth (Shukla et al., 2015), aiding the colonization on roots surface and enhancing the production of secondary metabolites (Palmqvist et al., 2015; Timmusk et al., 2018; Khan et al., 2018; Panichikhal et al., 2019). Nanocomposite formulations consisting of metal/metal oxide NMs and PGPR producing siderophores and lipopeptides, could be designed for improving the stability and dissolution of NMs as well as plant growth and health promotion. But how PGPR interact with NMs at the population level in the rhizosphere and the consequences on plant health are still unknown. Moreover, the “rhizosphere effect” among plant species is governed by the type and quantity of root exudates, leading to the differential microbial abundance and composition (Philippot et al., 2013). The stronger ‘rhizosphere effect’ of plant system may be more promising to engineer the bioavailability of NMs. Maximizing these potentials demand more deeper understanding of the *in planta* interactions between NMs and PGPR. Additionally, microbial activities could be inhibited by NMs. In the study of Simonin et al (2018), the microbial activities and processes (denitrification, nitrification, and soil respiration) were significantly reduced by 100 mg kg⁻¹ CuO NMs, however, CuO NMs exposures at relative low concentrations (0.1 and 1 mg kg⁻¹) had limited effects on soil microbial activities. Thus, in nano-enabled agriculture, appropriate concentration and application methods need to comprehensively consider to reduce environmental risks. Accordingly, biological risk assessment of NMs in agricultural applications should be conducted carefully before any large scale use in the future.

2.4. Soil fauna: Taking earthworms as an example

Soil fauna is commonly considered as terrestrial ecosystem engineers (Cunha et al., 2016), and plays an important role in soil processes, such as soil nutrient mineralization (Bardgett and Chan, 1999). Earthworms are a major representative group of soil fauna, strongly disturb and shape soil biotic and abiotic environments due to their large size and wide range of activities including feeding, digesting, and burrowing (Puga-Freitas and Blouin, 2015). The regulation of earthworms on geochemical behavior and bioavailability of NMs by rhizosphere processes can be divided into two steps: 1) Physical impacts of earthworm bioturbation on the NMs mobility from bulk soils to rhizosphere; and 2) direct and indirect impacts of earthworm mucus on NMs geochemical behaviors in rhizosphere (Fig. 4C). For step 1: to date, studies regarding the effects of earthworms on the mobility and availability of NMs in soils are still scanty. However, a column transport experiment with X-ray tomography showed that earthworm burrowing

regulated the migration and fate of nanoparticles in soils (Baccaro et al., 2019). As earlier studies suggest that earthworms contribute to the rapid transfer of pore water, air and solutes to horizontal and vertical soil layers, which in turn promote the mobility and availability of metals in soils (Jarvis 2007; Sizmur and Hodson, 2009; Leveque et al., 2014). Thus, the increased macroporosity and pore size caused by earthworm bioturbation can promote the transfer of NMs from bulk soils to the rhizosphere. For step 2: on one hand, as we discussed earlier, rhizosphere microorganisms play key role in driving the geochemical behavior of NMs in the rhizosphere. Numerous studies have shown that earthworms promote the abundance and activity of soil microbes (e.g., PGPR, plant symbiotic microbes) in rhizosphere, through their burrowing activity and secreting mucus (Cao et al., 2015; Braga et al., 2016), which in turn indirectly mediate the geochemical behavior of NMs driven by microbes in the rhizosphere. On the other hand, earthworms move in the soils accompanied by secreting mucus from their body surface, a mixture of water, electrolyte, indoleacetic acid and amino acid (Puga-Freitas and Blouin, 2015), which can enter the rhizosphere and interact with NMs. Actually, our present study indicates that the earthworm mucus notably affects the suspension, hydrodynamic size and solubility of metal oxide NMs (unpublished data). Similarly, previous study also found these various active groups of amino acids in mucus (e.g., carboxyl and amino) could accelerate the dissolution and reduction of NMs including CuO and CeO₂ NMs, and thus improve the NMs bioavailability (Gunawan et al., 2011; Odzak et al., 2014; Huang et al., 2017). However, the regulation of soil fauna, earthworms on the chemical behavior of NMs in rhizosphere is still in its infancy, this indeed calls for a better integration of earthworms into geochemical behavior of NMs driven by rhizosphere processes in future.

3. Biological responses of plants to NMs

After going through a series of rhizosphere processes, NMs might accelerate phytohormone biosynthesis, differential gene expression for elemental transporters, primary and secondary metabolisms, and antioxidant enzyme activity in the food plants. It has been shown that NMs regulate not only the absorption and transport of plant nutrients, but also the metabolic and biochemical pathways of nutritional quality (synthesis of carbohydrates, amino acids, and fatty acids) (Hatami et al., 2016). The antioxidant substances such as phenols, flavonoids and anthocyanins in licorice seedlings were increased by ZnO and CuO NMs (Oloumi et al., 2015). Upon graphene oxide exposure, phenylalanine metabolic pathway in rice was up-regulated (Zhou and Hu, 2017). 18 studies indicated that the nutritional properties of food plants were enhanced by NMs application (Table 1). After TiO₂ NMs exposure, the nutritional quality of rice grains in terms of P content was improved by the accumulation of P related metabolites (i.e., phosphoric acid and glycerol-3-phosphate) along with the enhancement of nitrogen sources such as glutamic and aspartic acid (Zahra et al., 2017). The root application of CuO NMs increased the content of soluble solids and the level of titrable acidity in tomato fruits, which might be due to that photosynthates were accumulated by the improvement of electron transport chain and phosphorylation during the light reaction, and the enzyme activity in the dark phase in response to NMs (Hernández et al., 2017). Zn-Amino nano-complexes could increase the phenolic compounds (catechin, hesperetin, and rosmarinic acid) in sweet basil, because Zn is a part of many enzymes as cofactor in plants (Tavallali et al., 2018). Among these studies, biological responses of plants upon NMs exposure play a crucial role in the enhancement effects (Fig. 5).

3.1. Signal transduction pathway of food plants in response to NMs

Fruit formation in plants is dependent on the biosynthesis of primary and secondary metabolites, which is regulated by the nutrients supply and signal transduction (Ruan et al., 2010). Signal molecules involved in the regulation of metabolite synthesis include abscisic acid

Table 1
Elevated nutritional quality in various food plant species by different NMs application.

Plant Species	NMs Type	Growth Condition	Exposure Method	Exposure Time (d)	NMs Modification	NMs Size (nm)	NMs Concentration	Unit	Edible part	Elevated Element	Elevated Nutritional Quality	Reference
Rice (<i>Oryza.sativa</i>)	TiO ₂	Soil	Root	105	Pristine	20	500, 750	mg kg ⁻¹	Grain	P	Amino acid, Sugar, Fatty acid, Vitamin, Organic acid	Zahra et al. (2017)
Rice (<i>Oryza.sativa</i>)	CeO ₂	Soil	Root	135	Pristine	8	500	mg kg ⁻¹	Grain	K, Ca, Na	Protein albumin, Total sugar, Nonreducing sugar	Rico et al. (2013)
Wheat (<i>Triticum.aestivum</i>)	CeO ₂	Soil	Root	94	Pristine	8	125, 250, 500	mg kg ⁻¹	Grain	P, K, Fe	Amino acid, Fatty acid	Rico et al. (2014)
Barley (<i>Hordeum.vulgare</i>)	CeO ₂	Soil	Root	169	Pristine	8	125, 250	mg kg ⁻¹	Grain	P, K, Ca, Mg, S, Cu, Fe, Zn, Mn	Amino acid, Fatty acid	Rico et al. (2015)
Barley (<i>Hordeum.vulgare</i>)	TiO ₂	Soil	Root	92	Pristine	24.1, 22.7	500, 1000	mg kg ⁻¹	Grain	P, Ca, Mg, Zn, Mn, Ni	Amino acid, Crude protein	Posčić et al. (2016)
Pea (<i>Pisum.sativum</i>)	ZnO	Soil	Root	65	Pristine Doped Coated	10 15 20	250, 1000	mg kg ⁻¹	Seed	P, Fe, Zn, Mn	Total sugar, Reducing sugar	Mukherjee et al. (2016)
Peppermint (<i>Mentha.piperita</i>)	TiO ₂	Soil	Foliar	150	Pristine	21	50, 100, 150, 200, 250	mg L ⁻¹	Leaf	N	Chlorophyll, Essential oil, Menthol, Menthone, Menthyl acetate	Ahmad et al. (2018)
Tomato (<i>Solanum.lycopersicum</i>)	Cu	Soil	Foliar	136	Pristine	50	50, 125, 250, 500	mg L ⁻¹	Fruit	K	Total proteins, Vitamin C, Total phenols, Flavonoid	López-Vargas et al. (2018)
Cucumber (<i>Cucumis.sativus</i>)	CeO ₂	Soil	Root	53	Pristine	10	400, 800	mg kg ⁻¹	Fruit	K, Ca, Mg, S, P, Fe, Mn, Zn	Reducing sugar, Nonreducing sugar, Total sugar, Starch, Protein albumin, Protein prolamin, Protein globulin, Protein glutelin	Zhao et al. (2014)
Cucumber (<i>Cucumis.sativus</i>)	ZnO	Soil	Root	150	Pristine	27	250, 500, 750	mg kg ⁻¹	Fruit	K, Mg, Fe, Mn, Zn, S, P, K	–	Servin et al. (2013)
Tomato (<i>Solanum.lycopersicum</i>)	TiO ₂	Soil	Foliar Root Foliar Root	66	Pristine	25	100, 500, 1000	mg kg ⁻¹	Fruit	–	Lycopene	Raliya et al. (2015)
Tomato (<i>Lycopersicon.esculentum</i>)	TiO ₂	Sand	Root	120	Pristine	<100	5, 10, 20, 40	mg L ⁻¹	Fruit	–	Amino acid, Total phenolics, Antioxidant capacity	Khan (2016)
Tomato (<i>Solanum.lycopersicum</i>)	Cu	Soil	Root	90	Pristine	25	0.02, 0.2, 2, 10	mg g ⁻¹ Cs-PVA hydrogel ⁻¹	Fruit	–	Lycopene, Antioxidant capacity	Hernández et al. (2017)
Cucumber (<i>Cucumis.sativus</i>)	Cu	Soil	Root	68	Pristine	40	200, 400, 800	mg kg ⁻¹	Fruit	–	Sugar, Organic acid, Amino acid, Fatty acid, Vitamin	Zhao et al. (2017)
Wheat (<i>Triticum.aestivum</i>)	CeO ₂	Soil	Root	64	Pristine	8	100, 400	mg kg ⁻¹	Grains	–	Protein, Total sugar, Reducing sugar, Rosmarinic acid	Du et al.(2015)
Cucumber (<i>Cucumis.sativus</i>)	Cu	Soil	Root	88	Pristine	40	200, 400, 800	mg kg ⁻¹	Fruit	Cu, Fe	–	Zhao et al. (2016b)
Sweet basil (<i>Ocimum.basilicum</i>)	Zn	Soil	Foliar	7	Coated	<50	2000	w/v	Leaf	–	Catechin, Hesperetin	Tavallali et al. (2018)
Muskmelon (<i>Cucumis.melo</i>)	γ-Fe ₂ O ₃ Fe ₃ O ₄	Soil	Root	35	Pristine	20	100, 200, 400	mg L ⁻¹	Fruit	–	Vitamin C, Total amino acid	Li et al.(2019)

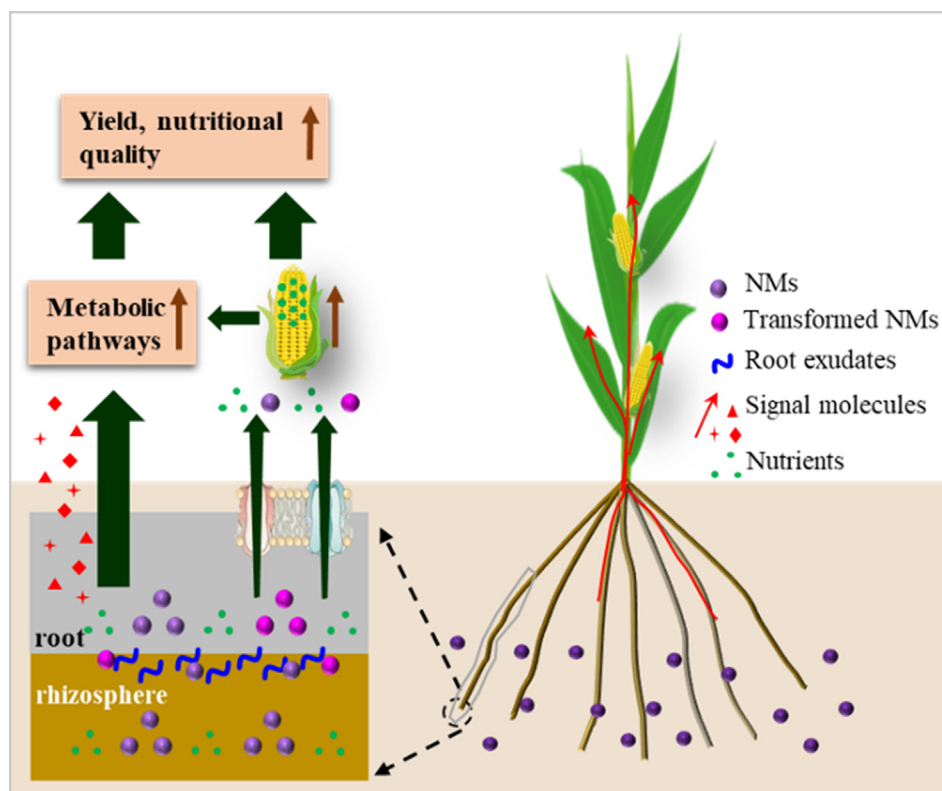


Fig. 5. Biological responses of NMs to enhance the yield and nutritional quality of food plants: 1) Signal transduction pathway of food plants in response to NMs; 2) promoted absorption and transport of nutrients by NMs.

(ABA), salicylic acid (SA) and jasmonic acid (JA). NMs can induce the biosynthesis of signal molecules, regulating the expression of defense genes, and synthesis of defensive secondary metabolites in plant seedlings, thereby affecting their resistance to biotic and abiotic stresses (García-Sánchez et al., 2015; Vankova et al., 2017). ABA and SA in *Arabidopsis thaliana* leaves and roots were significantly increased by ZnO NMs (Vankova et al., 2017). Similarly, ABA was significantly increased in cotton responding to CuO NMs exposure (Le Van et al., 2016). After La_2O_3 NMs exposure, the level of stress hormone ABA was increased in maize, and the core genes specific for biosynthesis of lignin were also increased, resulting in the increased lignin. But, the relative expressions of most aquaporin genes were downregulated, which led to the reduction of water uptake, subsequently causing significant growth inhibition (Yue et al., 2017; 2019). However, there is no study on the signal transduction in higher food plants induced by NMs during the whole life cycle, including the maturity period. Better understanding of the signal transduction in food plants in response to NMs could help regulate the biological responses and improve food plant yield and quality.

3.2. Promoted absorption and transport of nutrients by NMs

A 400 mg kg^{-1} ZnO and CeO_2 NMs elevated the Mg level in cucumber fruits due to CeO_2 NMs induced upregulation of Mg^{2+} transporters, e.g., aquaporin and Mitochondrial RNA splicing2 (MRS2) (Zhao et al., 2014). Hu et al. (2017) have shown that 50 mg L^{-1} Fe_2O_3 NMs could activated the expression of ferric-chelate reductase (FRO2) gene and iron-regulated ferrous iron transporter (IRT1) gene, promoting the absorption and transportation of Fe in *Citrus maxima*. Also, Se NMs elevated the expression levels of sulfate/selenate transporters (*Sultr1;2*), a main transporter for selenite/sulfate uptake into roots, *Sultr2;1* gene, which is responsible for selenite translocation to the shoot, and the ATP sulfurylases (ATPS) expression for Se hyperaccumulation. The elevated expression of these genes finally promoted Se uptake into edible plant

tissues (El-Ramady et al., 2016). Also, NMs might affect the genes annotated for groups responsive to endogenous stimuli such as hormones (Tumburu et al., 2017). Expression of *C. maxima* basic helix loop helix (*CmbHLH1*) transcription factor induced by ABA promoted the transport and distribution of Fe in *Arabidopsis* (Lei et al., 2014). SA could elevate auxin and ethylene signaling, thereby activating Fe translocation via the *bHLH38/39*-mediated transcriptional regulation of downstream Fe genes (Shen et al., 2016).

Above all, NMs can directly regulate biosynthesis of signal molecules in plants at the molecular level, mediate the transport of nutrients and expression of plant defense genes, and regulate a variety of secondary metabolic pathways. In order to regulate the production of proteins, sugars, and lipids in food plants, which are mediated by the nutrient absorption and transport, the mechanisms of NMs stimulating specific transporters need to be further explored.

4. Conclusion and future perspective

Surface coating could effectively enhance the bioavailability of NMs by alleviating the hetero-aggregation of NMs with soil particles and increase the interaction between NMs and plants. Root exudates could affect the physical processes (against the hetero-aggregation of NMs) and chemical processes (e.g. dissolution and oxidation/reduction) of NMs in soil. Rhizosphere microbes could improve the biological activity of NMs mediated by the process of dissolution and anti-aggregation. Earthworms also exhibited potential to increase the bioavailability of NMs by influencing soil physical, chemical and biological environment.

NMs could be delivered to root surface after increasing the bioavailability by surface coating, root exudates, and soil organism activities. NMs in plant cells can directly alter the biosynthesis of signal molecules, which could induce the gene expression of element transporter to promote plant nutrient uptake and secondary metabolism, thereby increasing the plant food nutritional quality. Therefore, there is great potential to enhance the mobility and bioavailability of NMs in

soils and promote biological response via managing rhizosphere processes in agro-ecosystem. However, research on the benefits of NMs to nutritional quality of food is still at an early stage. Considerable challenges limit the application of NMs in agriculture, and four such challenges are presented below:

1) Thus far, Ti, Cu, Zn-related NMs were reported to enhance the yield and nutrient quality. However, with their increasing application, metal ions will accumulate in the environment to threaten the ecosystem. The potential of environment friendly carbon NMs for the increase of food plant yield and nutrient quality needs to be explored.

2) Surface coating could enhance the efficacy of NMs applied in the soil. However, high concentration stabilizers were used for effectively enhance the bioavailability of NMs and the stabilizer have negative effect on the organisms. Therefore, need for more effective and environment friendly stabilizers is warranted.

3) Root exudates and rhizosphere organisms can alter the environmental behavior of NMs. However, the soil environment is complicated, field experiments should be carried out for better understanding the environmental behavior of NMs during agricultural application. The PGPR-NMs interaction and earthworm-plant-NMs interaction in rhizosphere seems more intriguing and may provide promising opportunity to explore novel nano-formulations for agricultural applications.

4) Plant disease and pest are the huge challenges for the food plant yield and nutritional quality. Using NMs as a new weapon for pathogen and pest control is another strategy for using nanotechnology in agriculture. Therefore, exploring new nano-composites that could be efficiently used for pathogen/pest suppression and nutrients supply will be an exciting area of study.

Conflicts of interest

The authors report no conflict of interest to this work

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Appendix A. Supplementary data

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